



Da Vinci Xi Surgical System In-Service Guide: OR Staff

Software Version: da Vinci OS4 v11

Welcome to Phase II of the da Vinci® Technology Training Pathway. This document is designed to be a facilitator guide for conducting hands-on training to OR staff during the da Vinci Xi® System Overview In-service. The in-service follows the operating room work flow that occurs with use of the da Vinci Xi system. If the staff have been previously trained on another da Vinci platform, reinforce that while da Vinci controls are similar between systems, setup and use vary due to the mechanics and new features of the da Vinci Xi system. For reference to the system user manual, go to www.intuitive.com. Updates may be provided electronically.

INTUITIVE

Participant Preparation

Determine if the participants have completed the recommended online training on Intuitive Learning as indicated on the da Vinci® Technology Training Pathway. Any activity should be demonstrated by the indicated participant. For discussions, tailor your facilitation style (Q&A vs. didactic) based on the participants' level of preparation. As the facilitator, read all notes, warnings and cautions.

In-Service Participant Roles	Participant Name
First assistant(s)	
Sterile OR staff	
Non-sterile staff	

Facilitator Instructions

The facilitator may play any role not represented by an in-service participant.

Equipment Preparation

✓	#	Da Vinci Xi Equipment
	1	Da Vinci Xi system
	1	Da Vinci Skills Model
	4	8 mm da Vinci Xi cannulas
	4	5 – 8 mm cannula seals
	4	5 - 12 mm Universal Seal
	1	Maryland bipolar forceps (training if available)
	1	8 mm bladeless optical obturator
	4	Patient cart arm drapes
	1	Column drape
	1	Instrument release kit

✓	#	Da Vinci Xi Equipment
	1	8 mm 0° endoscope
	1	8 mm 30° endoscope
	1	Monopolar curved scissor with tip cover
	1	Fenestrated bipolar forceps (training if available)
	1	12 mm 3rd party trocar
	1	Non-energy grasper (training if available)
	1	Da Vinci Xi monopolar instrument cable
	1	Da Vinci Xi bipolar instrument cables

✓	#	3rd Party Equipment
	1	USB flash drive
	1	Ruler
	1	Dry-erase marker
	1	Patient Neutral Pad
	1	8 mm blunt obturator
	1	Laparoscopic energy instrument

✓	Da Vinci Xi System Setup
	Power off da Vinci Xi system
	Disconnect blue system cables

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1 System Component Overview

Facilitator note Begin with a Q&A to allow participants to summarize key features and benefits of each major system component. Guide the discussion around value delivered by the features.

Learning objective

Summarize the top 3 features and benefits of each system component.

System component overview discussion



Key features of the surgeon console

3D viewer – enhanced vision of anatomy

Hand controls – intuitive motion translated into precise, fine, tremor filtered movements at instrument tips

Ergonomic design – increases ergonomic comfort for surgeon during surgery



Key features of the patient cart

Boom – adjustable support structure that promotes flexibility in positioning patient cart arms over surgical workspace

Universal arms – allow assignment and re-configuration of endoscope and instruments to desired ports during a surgical procedure, thereby enabling better tissue access and visualization

Helm touchpad – enables non-sterile OR staff to select patient cart setting and adjust boom height/position, improving system setup efficiency



Key features of the vision cart

Touchscreen monitor – provides high quality view of surgical image and allows non-sterile OR staff to adjust vision settings and troubleshoot faults

ERBE VIO® dV Generator – integrated with the system, this electrosurgical unit (ESU) indicates location of installed energy instruments and displays energy effect settings at the surgeon console, increasing surgeon autonomy

Endoscope controller – provides a high-intensity light source to illuminate surgical site and houses electronics that promote efficient endoscope setup

Intuitive Hub (optional) – A single platform that accesses Intuitive's digital ecosystem in the OR that allows capture of procedural audio/video, combined with da Vinci® data, to create content that can be used to collaborate with patients and peers

2 OR Setup, System Connections, and Startup

Facilitator note Check that the system components are powered down and blue system cables are disconnected after the previous activity.

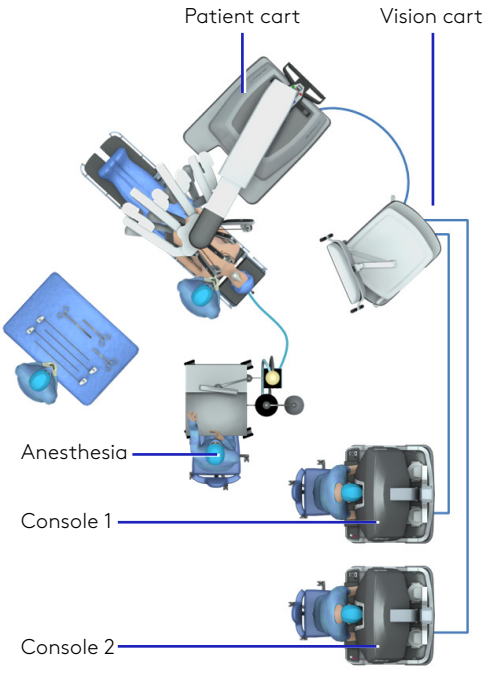
OR Setup

Learning objectives

- Articulate considerations for efficient da Vinci Xi® setup in the OR where training is held
- Position and setup system components in the OR
- Describe conditions under which system components are used in standalone mode
- Manage cables to maintain open traffic pathways, sterility and sufficient reach needed during surgery

OR setup considerations and standalone mode discussion

- Operating rooms may vary in size and shape. Consider access to doors and overhead structures.
- A universal room layout can decrease time required to "reset" the room between cases.
- An OR setup where the patient cart can be on either side of the patient allows the patient to be rotated 45° for assistant access on each side, minimizing movement of anesthesia.
- Determine power outlet locations (dedicated circuits) for each component to optimize OR setup.
- Each component can be powered on independently to perform specific tasks. Standalone mode is when the vision cart, patient cart and surgeon console are powered on and not connected to one another via the blue system cables.
- System cables are identical and interchangeable.



Note Specialty-specific setup is available in the da Vinci Clinical Specialty Guide.

Note If any individual component is in standalone mode and the blue system cables are connected between components, the remaining components will power on automatically to complete the integration. However, the integrated system must be powered off before disconnecting the blue system cables.

OR setup, system connections, and startup

Vision cart location activity

- Position the vision cart outside the sterile field on the same side as the anesthesia equipment or patient cart and ensure that it is:
  - Easily viewed by the patient-side assistant
  - Easily accessed by the circulating nurse
  - Close enough to the patient cart to allow unrestricted movement of endoscope and energy cables during surgery
- Check that the ERBE VIO® dV generator power cable (~9.5 feet long) can reach an AC power outlet on a dedicated circuit
- Check for reach of 3rd party cables such as patient neutral pad cables and insufflation tubing
- Practice managing system, power and auxiliary cables to maintain open traffic pathways
- Moving the vision cart:
  - Move the cart using only the handles and lock it in place by depressing the wheel brake switches

Vision cart standalone discussion

- Check that the blue system cables are disconnected from the vision cart core
- Power on the vision cart in standalone mode
- All vision cart functions are available in standalone mode
- Power off the vision cart

OR Application: Vision cart standalone mode can be used for training OR staff on touchscreen menus and settings that are accessed by non-sterile circulating staff during a da Vinci® case

Surgeon console location activity

- Position surgeon console outside the sterile field, in view of the operative field, in clear line of communication with the patient-side assistant
- Moving the surgeon console:
  - Use handles on the console sides ONLY
  - Press both brakes to lift up footswitch panel before moving, and press brakes again to lower the panel to the floor

OR setup, system connections, and startup

Surgeon console standalone mode discussion

1. Check that the blue system cables are disconnected from the surgeon console
2. Power on the surgeon console in standalone mode
3. A message appears on the touchpad that it is not connected to the vision cart
4. Power off the surgeon console

OR Application

- If a SimNow Simulator is available, it can be connected to the surgeon console via the short blue cable to power on the surgeon console in standalone simulator mode for skills practice
- Ergonomic adjustments can be made

Patient cart location discussion

- Discuss positioning the patient cart outside sterile field in an open space for startup to allow for movement of the arms during draping and to minimize chances of sterility breach
- During the procedure, the sterilely draped cart is positioned patient-side in the sterile operating field

Patient cart standalone mode activity

- Check that the blue system cable is disconnected from the patient cart
- Power on patient cart in standalone mode
- If possible, leave it connected to AC power to prevent draining the backup battery
  - The patient cart does not need to be connected to AC power for this mode
- All patient cart functions are available in standalone mode
- Press the cart drive enable switches on the handlebars, drive the patient cart within the OR
  - Use two people to move the cart, one to drive the cart and the other to provide directions using room landmarks
  - Manage system and power cables to maintain open traffic pathways
- Power off the patient cart

OR Application: Patient cart standalone mode can be used to transport the patient cart between operating rooms or over long distances (with sufficient battery power) in a hospital.

OR setup, system connections, and startup

System Connections

Learning objectives

- Integrate system components
- Connect 3rd party components to the system, if available

Power connections activity

- Confirm that vision cart, surgeon console and patient cart are plugged into AC power outlets that are on separate, dedicated electrical circuits
  - Keep patient cart plugged into AC outlet 24/7 to maintain battery power
- Plug ERBE VIO dV generator and auxiliary devices into separate dedicated circuit AC power outlets
- Vision cart shelves can be used to store auxiliary devices (insufflators etc.) up to a total of 50 lbs / 22.68 kg
- If the room is dedicated to da Vinci procedures, it is not necessary to unplug the power cables at the end of the day

System cable connections activity

1. Remove protective cap on each blue system cable and inspect connectors and system receptacles for debris
   a. Leave protective caps installed when not connected to the system
2. Attach one end of each cable to any receptacle in the vision cart core and the other ends to receptacles the rear of the surgeon console and patient cart
3. Align red marks on the cable connector and receptacle and push cable into position — an audible click confirms connection.

Note The short fiber cable plugged into Vision Cart core is the video processor's connection — do not remove this.

3rd party connections discussion

- Up to two video sources in one of the labeled formats can be plugged into any of the video input bays on vision cart or surgeon console, to enable auxiliary videos or images in the multi-image mode (TilePro)

  OR Application: Examples: View ultrasound images during surgery, or training videos during console skills practice

- Plug in video output devices into any of the video out bays on the surgeon console or vision cart, in labeled format

  OR Application: Use this for recording videos of surgical cases

- Use 'L' or 'R' labels on the video bay to select left or right eye transmission

  OR Application: The system allows 3D video recordings via the 'R' and 'L' video out bays

3rd party connections discussion continued

- Plug in RCA connectors of audio devices into the audio input or audio output bays

  OR Application: Use these bays for transmission of surgeon console audio during case recordings

- If using a 3rd party ESU, connect the system end of its cable into any of the 3 energy receptacles on vision cart core
  - The only approved 3rd party ESU is the Covidien ForceTriad
  - Be sure to stay within allowable maximum monopolar settings for the specific generator
  - Only use 3rd party generators that are validated for use with the surgical system to prevent unintended energy delivery to patient tissue

System Startup

Learning objectives

- Power on the integrated system and other 3rd party equipment
- Set up auxiliary and backup equipment

System startup activity

1. Check for interference of patient cart arms and surgeon console hand controls to prepare for startup
   a. Detach velcro straps on hand controls if they were left connected during transport
2. Press the power button on any system component to power on the system
3. Power on the system at the start of the day to allow it to go through the mechanical and electrical integrity tests and ensure proper function before patient is placed under anesthesia - it is not necessary to restart the system between cases
4. Observe mechanical integrity tests during startup: hand controls and patient cart arms move into position
5. Check that color bars are seen in right and left eyes of surgeon console viewer to confirm communication between the vision cart and surgeon console
6. Power on ERBE VIO dV generator
7. Power on 3rd party equipment
8. Confirm location of the backup equipment
9. Setup auxiliary equipment
   a. Install up to two carbon dioxide tanks in the tank holders for the insufflator

Learning check

1. What is standalone mode on the da Vinci Xi system?
2. What are the key considerations when setting up an OR for a da Vinci Xi case?

3 Vision Cart Overview

Learning objective

Interpret touchscreen interface and use it to complete typical OR setup and settings adjustment tasks.

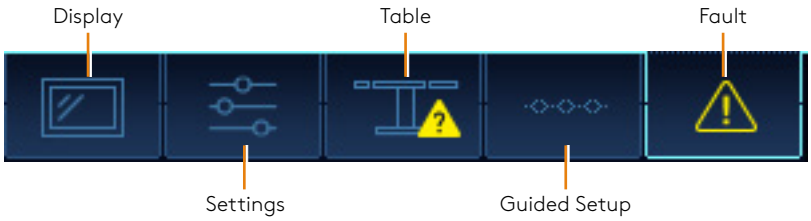
Touchscreen monitor discussion

- The touchscreen monitor is used to view the surgical field, view instrument and endoscope status and control vision system settings
  - The menu display area is always available on the right-hand side of the touchscreen monitor
  - Across the top is the system status and message area: non-critical messages can be hidden
  - Across the bottom of the screen are arm, instrument and endoscope status areas
    - This shows the name of the instrument installed, instrument arm message area, instrument activation status area, endoscope angle and orientation, horizon indicator, table status, Firefly® laser status and touchscreen zoom level
    - Messages relevant to the installed instruments and endoscope appear over the applicable arm status pods
- Main display shows the video feed received from the system electronics
  - When no video is being received, color bars will show on the main display
- For the vision system it is important to understand the safe and effective use of the vision system and its adjustment capabilities, to prevent deviation in workflow

Warning: Be aware that the messaging over pods indicates the arm associated with that pod; this is to prevent patient injury due to user confusion.

Touchscreen tabs discussion (menu display area)

- Display tab
- Settings tab
- Table tab
- Guided Setup tab (when applicable)
- Fault tab (when applicable)



Telestration activity

- Telestrate to communicate to the console surgeon from the vision cart touchscreen.
- Clear all telestration marks from the vision cart and surgeon console screen using the erase button or by pressing on the endoscope control pedal. The patient-side assistant can also erase telestration marks by pressing on the clutch button of the arm with the endoscope inserted.

Note To prevent loss of visualization, ensure proper setup and use of the vision system.

Vision cart overview

Display tab discussion

Display Tab	OR Application
Messages: Visible/Hidden	View non-critical system messages at both the touchscreen and 3D viewer
Take photo	Capture an image from the endoscope view when an endoscope is connected to the endoscope controller and a USB flash drive is connected to the video processor
Views:	
• Console view	View surgical image as seen in the surgeon console 3D viewer, including TilePro images and digital zoom levels
• Endoscope view	View endoscopic view of the surgical image
• TilePro 1/TilePro 2	View auxiliary images independent of surgical image
Left/Right eye	Select the left or right eye video image to identify vision or fogging issues in one eye channel or choose the telestration eye
Scope angle	Rotate the angle of the 30° scope up or down when a 30° scope is installed on an arm
Firefly On/Off	Activate or deactivate Firefly mode
Firefly intensity slider	Increase the contrast between the fluorescing tissue and non-fluorescing tissue
Turn Illuminator On/Off	Turn illumination on or off
Brightness	Adjust brightness of the surgical image
Erase telestration	Erase telestration marks

Vision cart overview

Settings tab discussion

Settings tab	OR Application
Uses remaining	View inventory overview of the instruments and accessories used during the procedure.
Video input/output	Specify video output format • 2D/3D Tilepro • Graphics off/on • 2D/3D video out • Connector type selection x3
Troubleshooting	
• 3D calibration	Perform manual 3D calibration of the endoscope. Not normally required.
• White balance	Perform manual white balance of the endoscope. Not normally required.
• Color balance	Adjust the color balance of the endoscopic image according to the surgeon's preference. • Sliding to the left increases the red tones in the image (e.g. when working near fibrous or avascular tissue). • Sliding to the right increases the blue tones in the image (e.g. when working near fatty or vascular tissue).
• Color modes *	The Color Mode features allow you to change the endoscopic image by adjusting the color and image characteristics. • **Standard color mode (default)** produces an image similar to open surgery. Tissue differentiation is more pronounced and blood appears redder than in classic color mode. • **Classic color mode** produces an image with a comfortable hue while keeping whites bright. It is ideal for looking at lower or constant contrast scenes (for example, fat, bowel, muscle). This mode closely matches the "Normal Color Mode" available on past Xi endoscopes. * Endoscope plus only - See Endoscope Plus Training Guide
• Color bars	Display a color bar image in either the left or right eye of the endoscope view. Used only for troubleshooting.
• Restore video defaults	Restore video settings back to their default values.
• Display event logs	Display the system event logs on the vision cart touchscreen.
• Onsite status	Displays the status of the network connection. The connection states are: disconnected, connected, or active session in progress.

Vision cart overview

Settings tab discussion continued

Settings Tab	OR Application
Troubleshooting	
• Software version	Lists the current system software version number.
• System identification	Lists the system name and serial numbers for the surgeon console, patient cart, and vision cart.
Patient cart mic	Mute or enable the patient cart microphone.
Volume	Adjust volume of surgeon audio heard on patient cart column speakers at request of patient-side staff.

Table tab discussion

- It is highlighted when table motion is available and grayed out when not available
- If table motion is available please reference the ITM In-Service Guide

Guided Setup tab discussion

- Provides the OR team with step-by-step instructions with audible voice prompts to successfully prepare, position and dock the patient cart to the patient
- When docking is completed, the Guided Setup tab disappears and is available again when all cannulas are undocked

Vision cart components discussion

Other components of the vision cart:

- Endoscope controller: Contains high intensity light source to illuminate the surgical site and electronics for initial processing of endoscopic video
- Video processor: Provides additional processing of endoscopic video and allows the recording of still images onto a USB flash drive
- System electronics (core): The system's central connection point where system cables and auxiliary equipment are routed

Learning check

1. What is the vision cart touchscreen used for? What does it display?
2. What are the tabs available in the menu display area?
3. What is Guided Setup?

4 Patient Cart Overview

Facilitator note During this activity, facilitate participant interaction with the boom and arms to build competency in these manipulations.

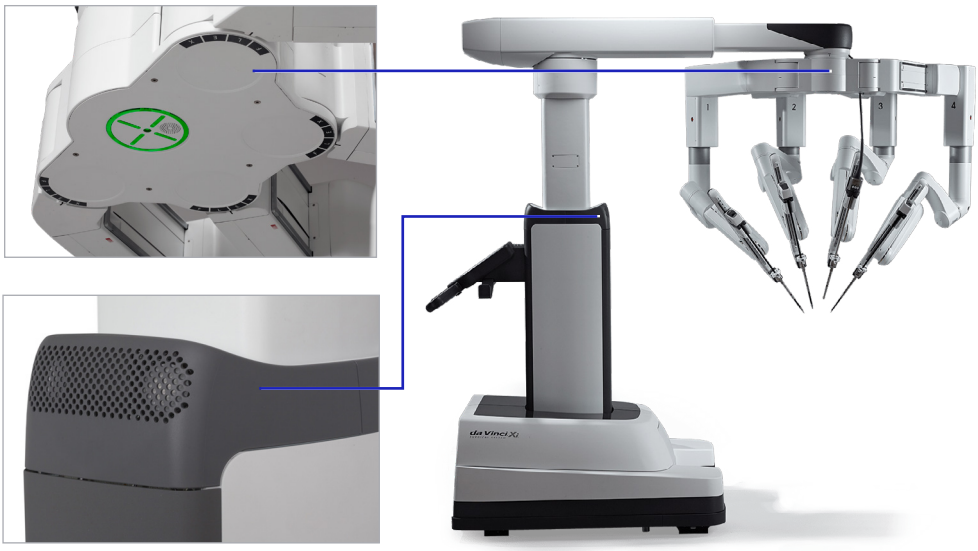
Patient Cart Arms Interaction

Learning objectives

- Identify patient cart components
- Manipulate arms accurately, with single button press and smooth arm movements
- Interpret boom and arm LEDs associated with arm joint activations

Patient cart components discussion

- Identify patient cart base, column, boom, patient-side speakers and patient-side microphone
- Identify arms numbered 1-4 from left to right: arms universally accept instruments or endoscopes
- Identify instrument carriage, insertion axis, cannula mount, and cannula mount lever



Patient cart overview

Patient cart arms activity



Patient Cart Arm	OR Application
• Clutch one arm - press and hold or quick click • Instrument clutch and carriage LED blinks blue	• Insert instruments under direct visualization • Adjust arm position without moving port/cannula
• Raise (▲) and lower (▼) the patient clearance joints on an arm using the patient clearance buttons • Instrument arm LEDs pulse blue	• Easily increase patient clearance as needed during surgery, without removing instruments • Increase internal instrument reach as needed beyond the level of ports, without having to remove instruments
• Grab and move arms individually • Instrument clutch, carriage and port clutch LEDs blink blue	• Make large movements to position undocked arms individually during draping • Move individual undocked arm up and away from patient at end of procedure or for conversion to open or laparoscopic surgery
• Port clutch one arm • Port clutch LEDs blinks blue	• Release tension at port sites • Adjust flex joints to manage arm spacing
• Cannula mount lever • Cannula mount	• Attach cannulas
• Raise and lower boom using extended port clutch - Press and hold the port clutch button on any arm and move it to its upper or lower limit - After the third beep, the arm will move toward the limit at which arm was held • Port clutch LEDs blink blue, boom LED blinks green	• Sterilely adjust boom height to avoid collisions or to access initial port for docking
• Rotate the boom using the boom rotation button on arm 1 or arm 4 • Boom LED blinks green	• Sterilely rotate the boom during mid-procedure re-dock to access a different target anatomy, without relocating the patient cart base

Patient cart overview

Patient cart LED discussion

- Solid blue: System is operating properly
- Blinking blue: Arm is clutched
- Pulsing blue: The arm is performing an activity which requires the user to wait for a brief period
- Blinking green: Guided Tool Change

Patient Cart Helm Interactions

Learning objectives

- Identify patient cart helm components
- Adjust boom height and voice prompt settings
- Make Guided Setup procedure selections

Patient cart helm components discussion

- Patient cart touchpad

  OR Application: Make Guided Setup selections, set boom height limit

- Power button and battery level indicator

  OR Application: Use the patient cart during a power emergency

- Boom position joystick

  OR Application: Manually adjust boom extension, pivot and rotation

- Boom height joystick

  OR Application: Manually adjust boom height

- Emergency STOP button

  OR Application: Stop da Vinci® surgery during an emergency

- Cart drive enable switches

  OR Application: Move and position the patient cart

Patient cart overview

Helm settings activity

1. Settings tab → Select Reduced to turn off non-critical Guided Setup voice prompts.

   OR Application: Reduced turns off tutorial voice prompts. If Reduced is selected, a message appears informing that tutorial voice prompts are turned off.

2. Move volume slider to adjust volume of patient cart speaker audio and voice prompts.

   OR Application: The patient cart speaker audio can be increased in a noisy OR to enable clear two-way communication between the console surgeon and patient-side staff.

3. Settings tab → Select On to turn on boom height limit.
4. Select Modify and follow on-screen instructions to modify boom height limit for the room where training is taking place. Select Save.

   OR Application: In operating rooms with low ceilings or overhead obstacles, setting a boom height limit will avoid external collisions when rolling up the patient cart to the patient to prevent injury to the user or patient.

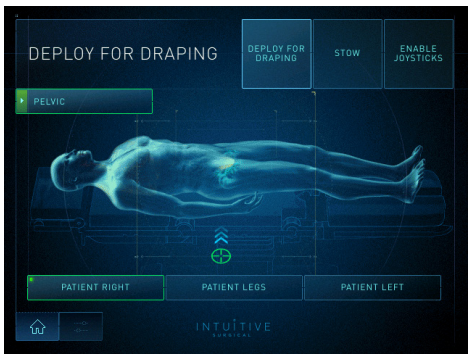
Guided Setup

- Provides step-by-step instructions with audible voice prompts, to successfully prepare, position and dock the patient cart to the patient.
- Instructions for the sterile surgeon and patient-side staff are displayed on the vision cart touchscreen. Instructions for the non-sterile circulating staff are shown on the helm touchpad.

Guided Setup activity

1. Select anatomy
2. Select patient cart location: side of the patient that the cart will approach from

OR Application: Select the side applicable for the current OR in which training is being held. Once selected, the touchpad image should match room setup.



Warning: Be sure to understand the function of the Patient Cart joystick and the use of the setup controls to prevent patient injury.

Learning check

1. When would you use each button on the patient cart arm?
2. What are the sub-components of the patient cart helm?

5 Draping

Facilitator notes

- During this activity, facilitate participant interaction with patient cart column and arm drapes to build competence in the draping process
- Provide guidance to participants to help avoid sterility breach when draping

Learning objectives

- Prepare the patient cart and sterile drapes for draping
- Follow Guided Setup prompts to practice draping the patient cart column and arms

Draping considerations discussion

- Follow proper drape installation and use of drapes, to prevent breach of sterility
- The column and all 4 arms of the patient cart must always be draped, even in 3 arm procedures, because all arms will enter the sterile field
- The column drape prevents contamination of the sterile, draped arms that will be positioned over the patient during surgery to prevent critical patient infection
- Each system requires 4 arm drapes and 1 column drape
- Draping requires both a sterile and a non-sterile OR staff

Warning: Drape all four arms for every procedure to prevent a critical patient infection.

Warning: Drape the Patient Cart column to prevent a critical patient infection.

Draping

Column draping activity

Non-sterile staff

1. On the helm touchpad, press and hold “Deploy for Draping”
    a. Guided Setup instructions are displayed on the helm touchpad and touchscreen monitor to guide the OR staff through the draping steps
2. Deliver column drape to the scrub nurse in a sterile fashion



Sterile staff

3. Read touchscreen Guided Setup message "install column and arm drapes"
4. Observe yellow undraped arms image
5. Position drape with drape card facing up
6. Release the first tear-away tab
7. Partially unfold drape on a sterile table
8. Open the flex strips to be at 90 degrees
9. Pick up the drape by holding the drape card within the drape cuff
10. Move toward the column, being careful to prevent cross-contamination
11. Attach the metal disc on the drape card to the magnetic socket on the column
12. Snap the tear-away tabs and unroll the remainder of the drape
13. Adjust the drape around the column as needed
14. Observe column image is no longer highlighted and appears blue
15. Adjust the drape around the column



Draping

Facilitator note

During the arm draping activity, encourage the sterile staff members to practice draping once with minimal assistance from a circulator, and once with help from a circulator to extend the drape to the back of the arm.

Arm draping activity



Non-sterile staff

1. Move each arm as needed to make sure there is enough space to drape each arm to prevent breach in sterility
2. Deliver the patient cart arm drapes to the sterile staff in a sterile fashion

Sterile staff

3. Orient drape with the tear-away tab facing the ceiling and sterility band up
4. Release the first tear-away tab
5. Partially unfold the drape on a sterile table so that the hand stickers face the ceiling
6. Extend excess drape upwards to reveal the arm clip
7. Flip the drape over so the instrument sterile adapter faces the ceiling
8. Grab, but do not break, the tear-away tabs along the drape cuff and spread to expose opening
9. Grab drape above sterile adapter and fold down to hold it with the drape cuff in the same hand
10. Tent the drape opening, holding the opposite end of the drape with the other hand
11. Carefully lower the drape over the patient cart arm
12. If needed, extend the arm using 'grab and move' to prevent contamination during installation
13. Attach the instrument sterile adapter
    a. Stabilizing the carriage with your hands, use both thumbs to click adapter into place
    b. Wait until you will hear an audible signal indicating sterile adapter engagement

    Note: Ensure that the drape pouch attached to the sterile adapter does not become pinched between the sterile adapter and the instrument carriage. If this occurs, remove the sterile adapter, straighten the drape pouch, and reseat the sterile adapter.
14. Observe the Guided Setup arm image is no longer highlighted and shows a blue check mark
    a. If the adapter does not engage, remove it by pinching the release tabs on the sides of the adapter, then reseat it
15. Place hands inside the drape cuff at the hand icons, with fingers behind the metal discs

Arm draping activity continued

Sterile staff





16. Orient hands at 6 o'clock and 12 o'clock positions
17. Move drape along the arm towards column
    a. The green band on the drape indicates the sterility barrier
    b. If a non-sterile staff member is assisting, they must not grasp the drape anywhere below the color band to prevent breach in sterility
18. Attach both metal discs on the inside of the drape cuff into the magnetic sockets on the arm ensuring that the flex-strips are facing the front of the arm
19. If needed, gather excess drape material and move it toward the back of the arm
20. Align the cannula mount sterile adapter and press straight in to click into place
21. Release the tear-away tab located near the arm clip on the back of the drape
22. Carefully slide excess drape material away from the gray handle and attach arm clip to the back of the arm to prevent patient infection or procedure delay
23. Move excess drape back to clear the sterile adapter for instrument insertion
24. Bend the flex strips in a "U" shape to create a clear instrument insertion path
25. Move the draped arm away from the undraped arms to avoid contamination
26. Drape remaining arms
27. Push arms behind laser line

Non-sterile staff

28. Press and hold "Sterile Stow" on the helm touchpad to compact the arm toward the column

Learning check

1. When is it necessary to drape arm 4?
2. What resources are required for draping the patient cart?

6 Instruments and Accessories Overview

Facilitator note Facilitate participant interaction with the cannulas, cannula seals, obturators, instruments and endoscopes.

Learning objective

Interact with and set up sterile accessories, endoscopes, and instruments for a da Vinci® procedure.

Accessories overview activity

Identify the following:

- Da Vinci cannulas
  - 8 mm reusable, available in 10 cm (standard) and 15 cm (long) lengths
  - Cannula fin
  - Remote center markings on both cannula lengths
    - The remote center is the pivot point around which the arm moves
    - Positioning remote center in the muscle layer of the body wall helps minimize the amount of force the system may exert on the body wall during surgery
- Cannula seals
  - 5-8 mm disposable seals
  - 5-12 mm Universal Seal
  - Universally used with both 8 mm instruments and endoscopes
  - All have inlet for use with insufflation or smoke evacuation as needed
  - Can also be used with 5 mm laparoscopic instruments
  - Place the cannula seal into the cannula bowl and push down until you hear an audible click
    - To remove seal, squeeze the latches towards each other and pull cannula seal out of cannula
- Da Vinci obturators
  - 8 mm blunt reusable in 10 cm and 15 cm lengths
  - 8 mm bladeless optical disposable in 10 cm and 15 cm lengths
  - Attach obturator (blunt or bladeless) to cannula assembly, ensuring that it snaps into the cannula with an audible click to indicate full engagement

Note For additional instructions on da Vinci bladeless obturator assembly, setup, and use, refer to the Instrument and Accessories User Manual.

Instruments and accessories overview

Accessories overview activity continued

- Da Vinci Hasson Cone
  - 8 mm (reusable)
  - 12 mm & stapler (reusable)



8 mm



12 mm

Note For additional instructions on da Vinci Hasson Cone assembly, setup, and use, refer to the Instrument and Accessories User Manual Addendum.

Note Use only instruments and accessories approved by Intuitive Surgical. System compatibility with non-approved instrumentation cannot be guaranteed. Improper assembly or disassembly of instruments or accessories can lead to tissue injuries.

Note For additional information about the EndoWrist Suction Irrigator In-Service Guide, reference.

Warning: Never use a double cannula (for example, an 8 mm da Vinci Xi cannula installed through the da Vinci Xi 12 mm & Stapler cannula) on the da Vinci Xi System or the da Vinci X System. This may cause instrument damage or an imbalance of the remote center, and potential injury to the patient.

Warning: Reprocessing or reuse of products intended for single use only may result in degraded instrument performance or loss of functionality, and in exposure to viral, bacterial, fungal, or prionic pathogens.

Caution: Da Vinci instruments must be used with the appropriate size and type Intuitive cannula, because tissue injury or instrument damage could result.

Instruments and accessories overview

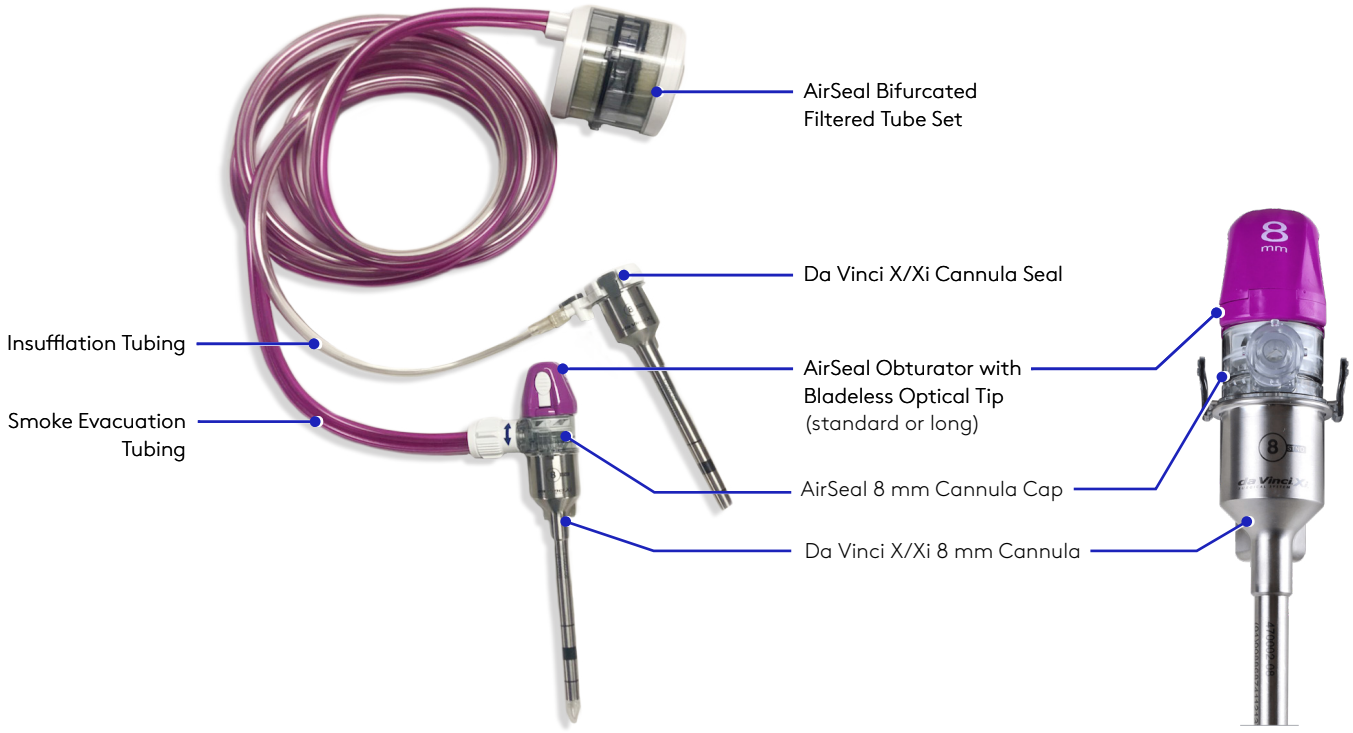
Third-party accessories discussion

The ConMed AirSeal® 8 mm cannula cap and obturator with bladeless optical tip comes in standard length and long length, and is compatible with the standard length da Vinci X/Xi 8 mm cannula and the long length da Vinci X/Xi 8mm cannula. The da Vinci X/Xi 8 mm blunt or bladeless obturator (optical) is not compatible with the AirSeal Cannula Cap.

The ConMed AirSeal Bifurcated Filtered Tube Set with Activated Charcoal Filter is compatible with the following da Vinci X/Xi Cannula seals:

- 5 - 8 mm cannula seal
- 12 mm & stapler cannula seal
- 5 - 12 mm Universal Seal

The image below represents a complete setup of ConMed AirSeal products with da Vinci X and Xi accessories. For additional instructions on ConMed AirSeal accessories, including assembly information, refer to the ConMed AirSeal Accessories user manual.



Caution: Unintended instrument tip motion can occur when the instrument carriage and accessories connected to the corresponding cannula bowl collide, resulting in patient injury.

Note Using third party accessories connected to the cannula may result in reduced range of motion of instruments due to interference between accessories and instrument carriage.

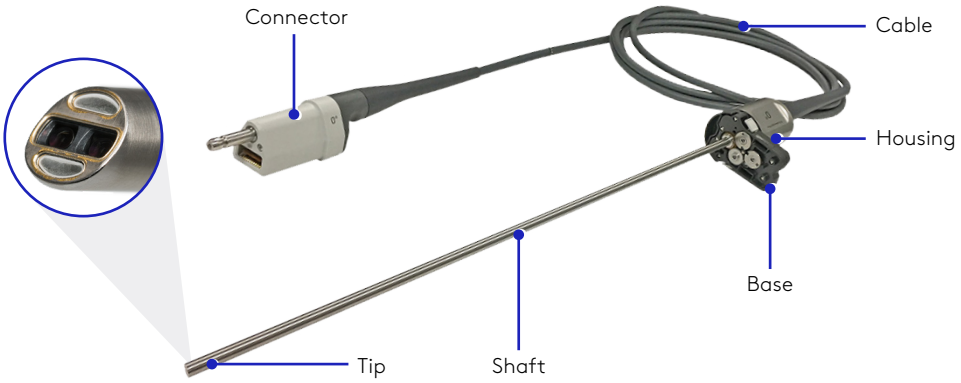
Instruments and accessories overview

Endoscope Plus overview activity

3D high definition endoscope

- For 0° and 30° endoscope, identify the following:
  - Integrated base, connector, housing, shaft and tip
  - Angled tip of 30° scope

See the Endoscope Plus Training Guide for information regarding physically differentiating endoscope plus from previous generation endoscopes



Note Inspect da Vinci Xi endoscopes (0° and the 30°) for deposits or damage to lenses, cable or integrated connector before and after each use.

Identify buttons on endoscope housing to prevent patient tissue injury from unintentionally turning off the endoscope during a procedure:

1. **Right/Left eye swap and targeting:** Identify vision/fogging issues in one eye of the endoscope or press and hold to activate the targeting feature
2. **Take Photo:** Capture a still image of the endoscope view
3. **Illuminator on/off:** Turn on the illuminator during laparoscopic use or turn off the illuminator sterilely upon endoscope removal from the patient cart arm



Warning: Avoid direct interaction with the illuminator to prevent burn injury or infrared radiation injury to the user.

Plug-in the 30° endoscope integrated connector to the endoscope controller on the vision cart.

OR Application

- In preparation for a da Vinci case, connect endoscope ~30 minutes prior to surgery
- Ensure endoscope is securely connected and ready for use

Note It is physically possible to install more than one endoscope on the patient cart arms. However, only the endoscope connected to the Endoscope Controller sends video to the 3D viewer and touchscreen monitor. If more than one endoscope is installed on the arms, a system message appears indicating that only one endoscope will send video.

Endoscope Plus overview activity continued

Identify Endoscope Controller status LEDs:

- Powering up (blue, flashing)
- Powered up and ready (blue, steady)
- Failure (red)
- No power (LED off)

Practice orienting the 0° and 30° scopes for laparoscopic use, with the base always facing up:

- For the 0° scope: buttons facing up
- For the 30° scope: align up or down arrow on the housing with the hash mark on the base
- Take care not to exert excess force on the endoscope shaft during laparoscopic use
  - Excessive force could cause bending of the endoscope shaft, especially when placed between two hard surfaces (ex: ribs, model)

OR Application: Da Vinci Xi® endoscopes can be used laparoscopically during port placement and as needed

Warning: Improper use or handling of any components that could get hot could lead to burn injuries.

EndoWrist instrument overview activity

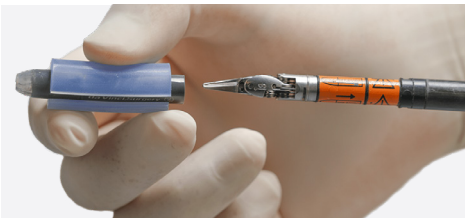
- Identify 8 mm da Vinci Xi EndoWrist® instrument components:
  - Tip, wrist, shaft, housing, release buttons, instrument release socket, maximum use indicator, flush ports, and discs
- Demonstrate/mimic the instrument's 7 degrees of freedom:
  - Insertion, external pitch, external yaw, roll, grip, internal pitch, and internal yaw
- Discuss instrument lives:
  - Most have 10-18 lives
  - Clip appliers have 100 fires
  - Training instruments have 30 lives or 100 fires
  - An instrument life is used as soon as the instrument goes into following mode from the surgeon console
  - If an instrument is expired, the maximum use indicator on the side of the housing will turn red

Monopolar curved scissors (MCS) setup activity

MCS tip cover

The tip cover is a single use accessory that is intended to provide insulation over a section of the monopolar curved scissors so that energy is only available at the tip.

- The MCS must always be used with the tip cover
- The tip cover should be inspected for damage if collisions occur during the procedure
- If the tip cover has been damaged in any way, remove the MCS to replace the tip cover, to prevent arcing through the defect and to prevent energy delivery to unintended tissue resulting in patient injury
- Improper application can lead to components coming off in the patient



Tip cover installation



Tip cover properly installed

Install the MCS tip cover following these steps:

1. Inspect for damage before installation – do not use a torn tip cover accessory
2. Close MCS blades and straighten the wrist
3. Grasp tip cover with blue installation tool and slide it onto the distal end until it stops
4. Remove installation tool and retain it
5. Check that the orange surface is fully covered, but the tip cover is not installed over the shaft
    a. The tip cover is not properly installed if any part of the orange surface is visible



Tip cover not properly installed



Tip cover over-installed

Warning: Do not use another instrument to clean the Tip Cover Accessory to prevent energy delivery to unintended tissue, resulting in patient injury.

Learning check

1. How do you orient the 30° and 0° endoscopes laparoscopically?
2. How many lives do most EndoWrist® instruments have?

7 Port Placement and Docking

Facilitator notes

- Demonstrate port placement and facilitate participants' practice of placing ports for multiple surgical workspaces
- Have participants practice steps specific to their roles in the OR. Play the other roles as required.

Port Placement

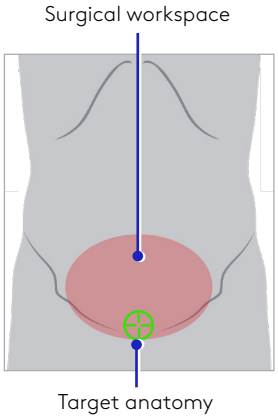
Learning objectives

- Describe port placement philosophy and general guidelines to follow when placing ports for a da Vinci® case
- Practice placing ports for example surgical workspaces relevant to the surgical specialties that the participants support
- Follow Guided Setup for preparing, positioning, and docking the patient cart
- Setup for a 3-Arm procedure

Surgical workspace activity

Define surgical workspace

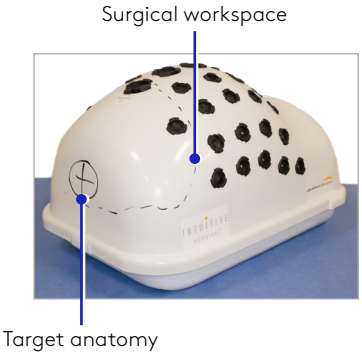
1. Area in the patient that all instruments will need to reach to successfully complete the procedure
2. Mark on the da Vinci Skills Model for a representative procedure



Target anatomy activity

Define Target Anatomy

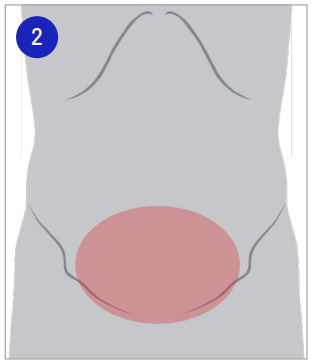
1. From the center of the surgical workspace, the target anatomy is the farthest point required to reach within the workspace
   a. This is not necessarily the location of pathology
2. Mark on the da Vinci Skills Model for a representative procedure

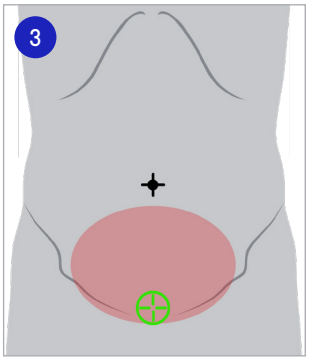


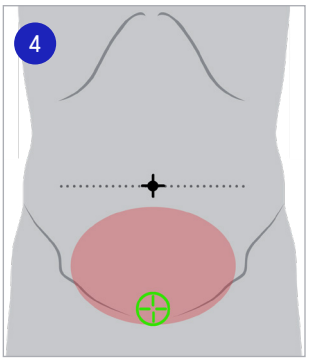
Port placement and docking

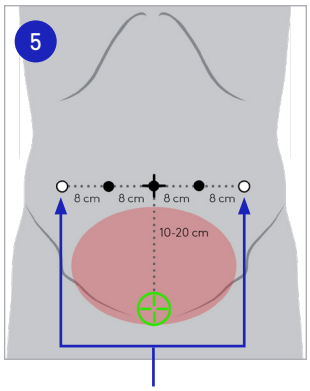
Port marking activity

1. Before marking port locations, insufflate the abdomen.
2. Identify the surgical workspace:
   a. If two quadrants or less, proceed to identifying the target anatomy.
   b. If greater than two quadrants, consider dual docking.
3. Identify a pelvic target anatomy, and mark the endoscope port 10 – 20 cm from target anatomy.
   a. If the workspace comprises two quadrants or less, the endoscope port will be at or beyond the edge of the workspace, ensuring adequate visualization.
   b. If the desired port location is greater than 20 cm from the target, consider dual docking.
4. A linear port placement is recommended to minimize arm-to-arm interferences.
5. For instrument arm ports, measure from the initial endoscope port perpendicular to the axis between the target anatomy and the initial endoscope port.
   a. Mark ports 6 – 10 cm (8 cm recommended) from initial endoscope port in a line.
      - Ports can be a minimum of 4 cm apart in certain cases (e.g. procedures utilizing a small surgical workspace and requiring minimal outer arm motion, such as mitral valve repair).
   b. Maintain 2 cm or more between ports and any bony prominences.
   c. Do not place ports between other ports and the target anatomy.
6. Mark the retraction arm port on either side of the patient, 6 – 10 cm (8 cm recommended) from the nearest da Vinci port, and in line with all other da Vinci ports.
   a. Consider location of internal anatomy to be retracted.
   b. Consider placing on same side as monopolar instrument (allows grasper use in both hands).
   c. Consider placing arm opposite of the assistant.



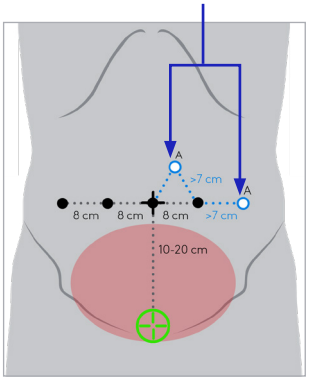






Retraction port options

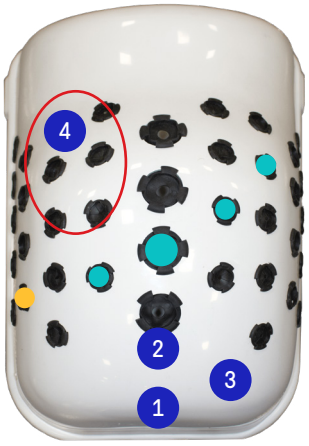
Assistant port options



Port placement and docking

Port marking activity continued

7. Consider port hopping potential when placing ports.
   a. Placing 4 ports allows switching between 2 right hands/left hands, changing vantage point.
   b. Placing 5 ports allows switching between two right hands and two left hands, while maintaining vantage point. The additional port can be used as an assistant port.
8. Mark the assistant port lateral to or triangulated between the da Vinci ports. Ensure the assistant port is at least 7 cm away from all other ports and has a clear path to the target anatomy.
   a. Use long-length laparoscopic instruments for greater reach and access.
   b. Do not place the assistant port between da Vinci ports and the target anatomy.
9. Mark ports for a second representative procedure relevant to your surgical specialty.
   a. For an upper right quadrant procedure, mark port placement as shown. The red outline represents the surgical workspace, and "4" represents the target anatomy.

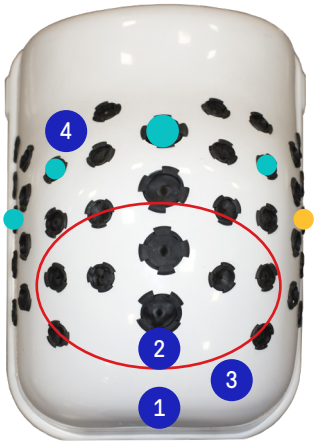


Port location:
upper right quadrant procedure

Port placement activity

If needed, re-mark ports for a pelvic procedure. Using the endoscope to provide direct visualization, place the instrument and accessory ports into the abdominal model as shown. The red outline represents the surgical workspace, and "2" represents the target anatomy.

1. Insert the endoscope port into the previously marked location
2. Insert the endoscope, and visually confirm the surgical workspace and target anatomy before placing additional instrument ports
3. Use the endoscope to provide direct visualization for placing the remaining ports
   a. Align the remote center of each cannula with the inside edge of the abdominal model, corresponding to the muscle layer of the abdominal wall
4. Insert the remaining instrument and accessory ports in the locations marked



Port location:
pelvic procedure

Patient positioning discussion

- Position the patient for a representative procedure at surgeon's discretion, following hospital protocol
- Once the patient cart is docked to cannulas in the patient, the patient cannot be repositioned

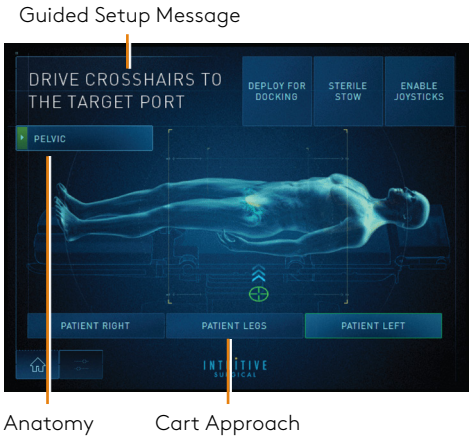
Docking

Guided Setup activity: Deploy for docking

1. Visually verify the anatomy and patient cart location selections on the vision cart touchscreen Guided Setup tab
2. Boom deployment position varies based on the anatomy and patient cart location selected at the helm
3. Ensure there are no obstructions overhead or in front of the patient cart
4. Following Guided Setup messages on the vision cart touchscreen and audible voice prompts, guide circulating staff to deploy the boom for docking
    a. Monitor boom movement to avoid collision with equipment or the patient
5. The boom will raise, extend, and pivot, orienting the patient cart arms based on the anatomy and approach that were previously selected at the patient cart helm

Guided Setup activity: Drive laser lines to the endoscope port

Using room landmarks for directions instead of “left” or “right”, direct the circulating staff to drive the patient cart towards the patient until the green positioning laser lines projected on to the floor are within 5 cm of the initial endoscope port.



Port placement and docking

Guided Setup activity: Dock the arm you will use for the endoscope

Select the arm that will act as the initial endoscope arm for the procedure and dock it to the port that will initially be used for the endoscope:

1. Lower the arm down to the cannula using the port clutch button
2. Angle cannula toward the cannula mount
3. Hold down the cannula mount lever with thumb while holding down the port clutch button
4. Insert the cannula fin into the cannula mount and release the cannula mount lever
5. When the cannula is properly seated, the lever will return to the upright position
    a. If it is not properly seated, the lever will remain down and a message will appear on the touchscreen over the associated arm pod
    b. Once a cannula is docked to an arm, the helm touchpad will indicate, "Surgery in Progress," show anatomy and cart location selections, and disable helm positioning functions
    c. Patient cart stabilization feet lower, locking arms into place

Warning: Do not move the Patient Cart while one or more arms are docked to the patient, to prevent harm to the user or patient.

Guided Setup activity: Install the scope

1. Insert the 30° endoscope into one of the undocked ports to check remote center of the initial endoscope port and adjust as necessary
    a. Ensuring correct placement of the remote center before docking an arm helps minimize the amount of force the system may exert on the body wall during surgery
2. Install the 30° endoscope onto the arm
    a. The endoscope will rotate to the position in which it was last used laparoscopically, to maintain the vantage point
    b. Do not place the endoscope cable in any pinch points
3. Secure endoscope cable between endoscope and insertion axis of the arm
4. Place endoscope tip in cannula, then install the base onto the sterile adapter until fully seated
5. Observe that the illuminator turns on automatically upon installation



Correctly secured endoscope cable

Port placement and docking

Facilitator note

To facilitate incomplete targeting activities, set a boom height limit of 90", and increase table height. Instruct participants to attempt targeting to complete the below activity. After practice, ensure that targeting is successful before proceeding with the remainder of Guided Setup.

Guided Setup activity: targeting

1. Quick click the instrument clutch button, and insert the endoscope into the Skills Drills Model.
2. Point the endoscope at the target anatomy, and lock the endoscope in place with a quick click of the instrument clutch button.
   a. Level the horizon of the endoscope to ensure proper boom orientation during targeting.
3. The targeting button is used to orient and align the patient cart boom to the center of the surgical workspace. Aligning the initial endoscope arm with the target anatomy enables optimal orientation of all patient cart arms over the surgical workspace.
4. Use one hand to stabilize the cannula remote center at the model's body wall to prevent patient injury. With the other hand, press and hold the targeting button.
5. If targeting does not complete, remove the endoscope and undock the arm. Attempt one of the following to troubleshoot incomplete targeting:
   a. If possible, follow the vision cart touchscreen message to "lower the operating table for improved sterile clearance."
   b. Raise or disable the boom height limit from the Settings tab of the helm touchpad.
6. Reattempt targeting, and observe the following:
   a. The boom will rotate to align with the principal working direction.
   b. The boom will pivot to center over the initial endoscope port.
   c. The boom height will adjust based on the height of the docked arm to ensure that all arms will be able to reach
   the remaining cannulas, as well as provide adequate clearance between the top of the patient cart arms and
   the boom.
7. After targeting completes, use the positioning laser lines as a guide to confirm that the laser lines transect the ports. These lines remain on for a brief period of time after targeting has completed, or until a second cannula is docked.
8. Using the port clutch button, flex the endoscope arm to bring it in line with the center of the workspace (i.e. along the positioning laser that points towards target anatomy).
   a. When using the port clutch button to reposition an arm, always stabilize the cannula at the body wall.

Warning: Be sure to understand the function of the targeting feature, including supporting the cannula while targeting, to prevent patient or user injury.

Manual boom orientation discussion

If targeting does not complete after lowering the operating table and raising the boom height, manually position the boom and arms using the following guidelines:

- Rotate the boom towards the target anatomy
- Ensure all arms can reach cannulas for docking
- Ensure there is enough space between the boom and arms to remove instruments in a sterile fashion

Warning: Undock the instrument arms prior to moving the patient, table, or Patient Cart to prevent patient harm.

Facilitator note

To facilitate the remaining docking activities, ensure that targeting is successful before continuing.

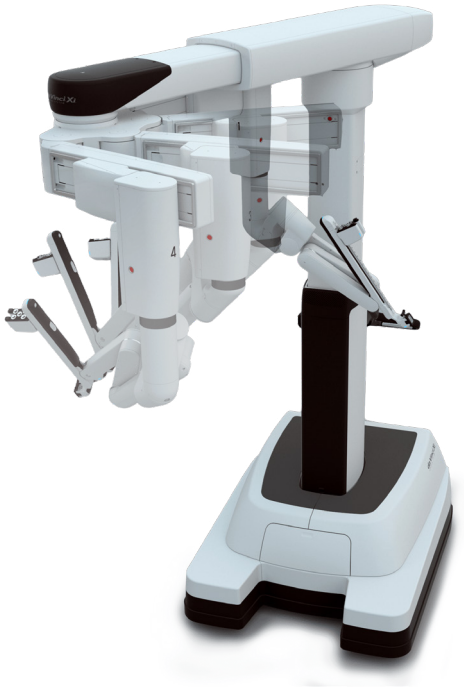
Guided Setup activity: Docking and system alignment

1. Dock the remaining arms as previously instructed
2. After the last arm is docked, the Guided Setup tab disappears
3. Arm spacing is adjusted to maintain parallel configuration of the arms because:
   a. Insertion axes of the da Vinci Xi® arms do not telescope
   b. External and internal arm-to-arm interferences are minimized in a parallel setup
4. Starting from the arm closest to the initial endoscope arm, flex the remaining arms towards the initial endoscope arm, maintaining about a fist's space between them
5. Check system setup spacing and ensure there is no tension at any port
   a. To release tension, stabilize the cannula at the body wall with one hand and press the port clutch button with the other hand
6. If maximum instrument reach is anticipated in a large surgical workspace, lower the patient clearance joint to about a fist's space to the patient
   a. This joint of each arm can be adjusted for either patient clearance or instrument reach

Port placement and docking

3-Arm Procedure Setup Activity

1. Drape all 4 patient cart arms and dock the arms to be used for the procedure, including performing targeting
2. Collapse the arm to be stowed onto itself using the instrument clutch button
3. Use the port clutch button to fully rotate the arm inward towards the other arms, until it is facing the column in the opposite direction of the other arms
4. Raise the arm to its upper most limit
5. Check that LEDs on the arm turn off and the 3D viewer displays "arm stowed"
6. Once the procedure is over, be sure to un-stow the arm and move it back to its normal orientation before moving the boom



Learning check

1. Define target anatomy and surgical workspace in the context of a da Vinci Xi procedure.
2. What purpose does targeting achieve?

8 ERBE VIO dV Generator

Facilitator notes

- Confirm that the ERBE VIO® dV generator is powered on
- Connect patient neutral pad

Learning objectives

Interpret and interact with the ERBE VIO dV (VIO dV) generator user interface and complete typical electrosurgical unit (ESU) setup tasks:

- Adjust energy settings and apply presets
- Set up da Vinci® and handheld monopolar and bipolar instruments for use with the ERBE VIO dV generator
- Navigate error logs for reference and troubleshooting

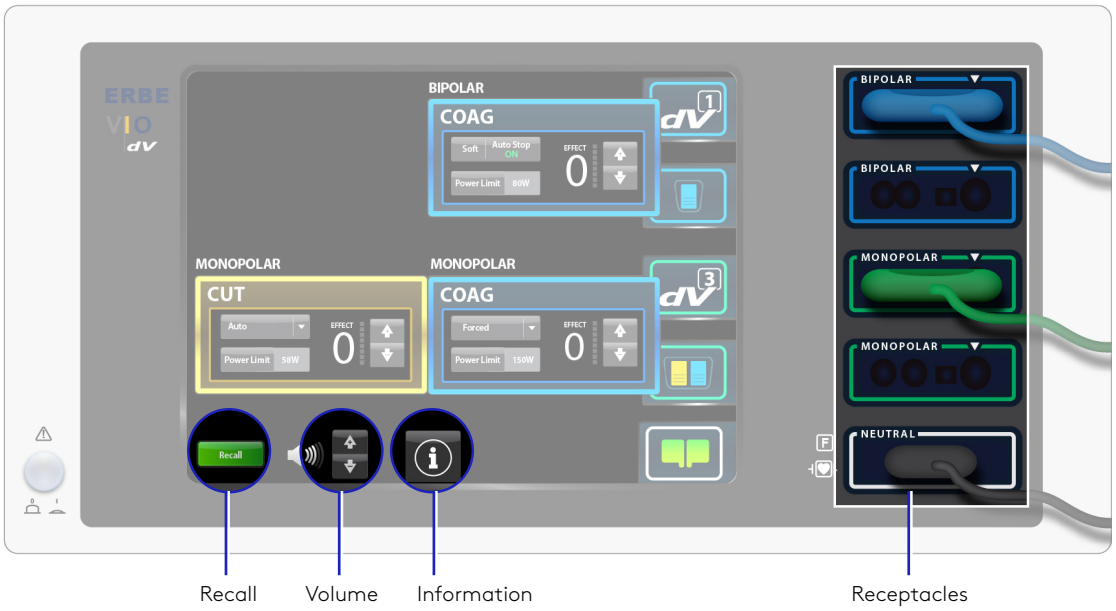
VIO dV overview activity

The VIO dV is an integrated Electrosurgical Unit (ESU) that can be used with da Vinci Xi® and handheld instruments.

Caution: Ensure that the energy is delivered from the intended generator to the intended instrument, to prevent patient or user injury.

Interact with the VIO dV generator to complete the following:

- Identify receptacles for da Vinci instruments and handheld instruments
- Press the "recall" button to recall previously applied settings
- Adjust the volume of audible feedback
- Open the "i" (information) menu, scroll down and review the characteristics and effect level range for each mode
- Close the menu when finished



ERBE VIO dV generator

VIO dV connections and use discussion





- The cautery cords are color coded to match the appropriate receptacles: Bipolar cords are blue and monopolar cords are green.
- Cautery cord generator connectors:
  - Each generator connector is molded on the top in order to fit the user's thumb when held in the correct orientation.
  - There is an arrow on the top of each generator connector that aligns the arrow on the top of each generator receptacle when installed correctly.
- Monopolar instrument connectors will only fit into the monopolar instrument cautery cord interface.
  - If a laparoscopic monopolar instrument needs to be used, it can be plugged into the single socket that is available on the instrument connector of the green cord. The generator-end of the green cord should be plugged into the bottom monopolar receptacle on the generator.
- Bipolar instrument connectors will only fit into the bipolar instrument cautery cord interface.
  - If a laparoscopic bipolar instrument needs to be used, it can be plugged into the two side-by-side sockets that are available on the instrument connector of the blue cord. The generator-end of the blue cord should be plugged into the bottom bipolar receptacle on the generator.

Note: When the bipolar cord is plugged in upside down, the instrument cannot be activated by the da Vinci system.

Note: Ensure instruments are connected properly to prevent delivery of energy to unintended patient tissue.

Laparoscopic instrument set up discussion

- Foot pedals for use with laparoscopic instruments are stored in the bottom drawer of the vision cart
- Foot pedals are hard wired for use with the bottom bipolar receptacle and the bottom monopolar receptacle which can be seen on the assignment indicators to the right of the mode pods
- If a laparoscopic instrument requires foot pedal activation, ensure that it is plugged into the correct receptacle. If the finger switch monopolar instrument is activated, it can be plugged into any of the available receptacle.
- Activation of da Vinci instruments by the auxiliary foot pedals is not allowed even if they are installed on the patient cart

ERBE VIO dV generator

Laparoscopic instrument set up activity

1. Connect the laparoscopic monopolar instrument to the single socket on the green cautery cord's instrument-connector
2. Plug in the generator-end of the cord into the second monopolar receptacle
3. Similarly, plug in the laparoscopic bipolar instrument into side-by-side sockets on the blue cord's instrument connector
4. Plug in the generator-end of the blue cord into the second bipolar receptacle

ERBE VIO dV setup activity

Facilitator note Perform instrument installations to support activities in this section. Participants will learn how to install instruments later in the in-service.

Setup a da Vinci bipolar instrument for activation:

1. Plug the blue cautery cord into the VIO dV generator such that arrow on the generator connector aligns the arrow on the bipolar receptacle
2. Plug the instrument-connector into its interface on the instrument housing
3. Install the bipolar instrument on arm 1. Observe the "?" icon change to number "1"
   a. The icon indicates that the generator detects the presence of a da Vinci Xi EndoWrist® instrument



Setup a da Vinci monopolar instrument:

1. Ensure a dual surface patient neutral pad is connected
2. Plug the green cautery cord into the VIO dV generator such that arrows on the connector and monopolar receptacle align
3. Plug the instrument connector into its interface on the instrument housing
4. Install the monopolar instrument on arm 3. Observe the "?" icon change to number "3"
   a. The icon indicates that the generator detects the presence of a da Vinci Xi EndoWrist instrument
5. Disconnect generator-end of the bipolar cautery cord, leaving the instrument end connected to installed instrument
6. Observe number "1" on the icon change to "?" for both bipolar receptacles
7. Reconnect generator-end of the bipolar cautery cord and uninstall both instruments from the system for the next section



ERBE VIO dV generator

Mode and settings discussion

There are two receptacles for bipolar instruments which are color coded blue.

- Bipolar mode pod
  - Controls the settings for both bipolar receptacles
  - The mode used for bipolar is "Soft Coag"
  - An effect of 1– 8 can be selected for "Soft Coag" as designated by the surgeon
  - Auto Stop can be turned on or off
    - When Auto Stop is on, the generator stops applying energy to prevent tissue from sticking to the instrument jaws

There are two receptacles for monopolar instruments which are color coded green.

- Monopolar mode pods
  - Monopolar mode pods will be grayed out until an appropriate patient neutral pad is attached, but the settings may still be adjusted
  - Plug in the patient neutral pad
    - The neutral electrode status will turn green and the monopolar mode pod will become active (For this in-service, partially fold the pad in half until the status turns green)
  - Controls the settings for cut and coag on both monopolar receptacles
- Cut Modes
  - Auto Cut: Effect can be adjusted between 1 – 8 as designated by the surgeon. Auto Cut is the default mode.
  - Dry Cut: Effect can be adjusted between 1 – 8 as designated by the surgeon
- Coag Modes
  - Swift Coag: Effect can be adjusted between 1 – 8 as designated by the surgeon
    - Swift Coag is the default mode
  - Forced Coag: Effect can be adjusted between 1 – 4 as designated by the surgeon
  - Classic Coag: Effect can be adjusted between 1– 2 as designated by the surgeon

Practice plugging in the patient neutral pad, selecting mode and adjusting effect.

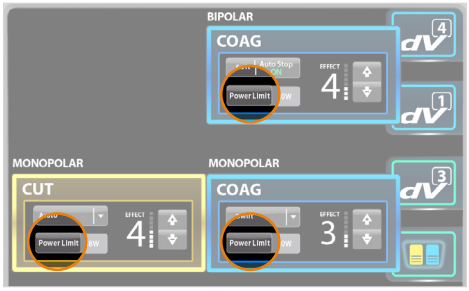
ERBE VIO dV generator

Power limit discussion

- Power Limit can be used to adjust the maximum power delivered for a set effect level, as designated by the console surgeon
- Adjusting the power limit enables more control of smoke, char, cutting and coagulation for all monopolar and bipolar instruments

Power limit activity

1. Select any “Power Limit” button
2. Touch the “Adjust Limit” button
3. Touch the left/right arrows to make incremental adjustments
    a. Touching the right arrow adjusts the power limit by increments of one
4. Touch “Default” to reset to the factory default power limit
5. Touch “Close” to return to the main screen



Error log activity

Error log tab displays the 20 most recent error messages. This log can be used for quick reference and troubleshooting.

1. Touch the “i” button located at the bottom of the VIO dV touchscreen
2. Confirm that the “Error Log” tab is displayed
3. Touch arrow button on the left next to an error message
4. Confirm additional information is displayed
5. Touch “Back” to return to the main screen



ERBE VIO dV generator

Energy presets discussion

- Surgeons can create and save up to three sets of preferred energy settings to their user account, from the surgeon console touchpad
- Upon request by the surgeon, these energy presets can be applied from the vision cart touchscreen by non-sterile OR staff
- Presets can be given custom names, and can be set to be applied by default at login

Apply presets activity

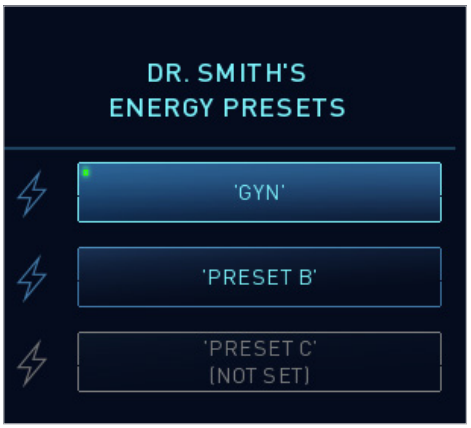
From the vision cart touchscreen, apply presets as follows:

In a **single console** setup:

1. Touch Settings tab
2. Touch "Energy Preset" button
3. Touch a preset from the preset list

In a **dual console** setup:

1. Touch Settings tab
2. Touch a user's name to display that user's presets
   a. If no preset is applied, the button will read "No Preset Selected"
3. Touch a preset from the preset list



Single console



Dual console

Learning check

1. Describe how you would setup a da Vinci bipolar instrument for use with the ERBE VIO dV generator.
2. What is power limit?

9 Intraoperative Setup and Use

Learning objectives

- Practice insertion, removal and exchanging of endoscopes and instruments
- Manage and troubleshoot endoscope fogging
- To prevent breach in sterility practice handling the instruments, endoscopes and accessories

Endoscope fogging discussion

- Technique for intraoperative cleaning:
  - Remove the endoscope from the arm
  - Wipe the tip with moistened sterile gauze
  - Briefly submerge the endoscope tip in warm water: temperature less than 55°C (131°F)
  - Wipe both the tip and shaft dry with sterile gauze
  - Use sterile gauze to clean the lumen of the cannula seal prior to installing the endoscope
- When the endoscope is outside the patient and waiting to be used, it can be placed in a scope warming device
  - Do not use warming devices that apply temperatures greater than 49°C (120°F) for extended periods of time (longer than 5 minutes)
- Use heated insufflation and check that insufflation/smoke evacuation tubing is not connected to endoscope cannula

Instrument insertion activity

For this activity, install a bipolar instrument in arm 1, MCS in arm 3, and grasper in arm 4.

1. Before inserting an instrument, clutch the endoscope and view the cannula in which the instrument will be installed to check and adjust remote center position
2. Straighten the instrument wrist and close the instrument tips
3. When using energy instruments, attach energy cables to the instruments before installing onto patient cart arm
4. Place instrument tip in the cannula then install housing onto sterile adapter until fully seated
5. Observe audible feedback and blue pulsing carriage LED during information download
6. Solid blue carriage LED and an audible tone indicate installation completion
7. Check icons on the assistant monitor to ensure that the instrument is engaged
8. Quick click the instrument arm and endoscope arm clutch buttons (blinking blue carriage LEDs)
   a. Observe off-screen indicator with associated arm number on the vision cart touchscreen
   b. Yellow and black for active instruments
   c. Gray and black for inactive instruments

Instrument insertion activity continued

9. Visualizing the cannula tip with the 30° scope, manipulate both arms to introduce instrument until its tip is just visible, then guide it towards the target anatomy
10. Quick click the instrument arm clutch button again to lock the instrument into place
    a. Do not move any instrument that is not under direct visualization
11. To prevent tissue injury, ensure that there is adequate room for instruments to move inside the patient, as unexpected motion can occur causing instruments to collide

Practice instrument insertion with a facilitator or another in-service participant manipulating the endoscope.

Practice instrument insertion independently manipulating both endoscope and instrument arms.

Instrument removal activity

Warning: Use proper instrument removal techniques (for example, straightening instrument wrist, freeing instrument jaws of patient anatomy) to avoid damaging tissue or the instrument upon removal.

Practice read-back technique between console surgeon and patient side assistant to remove an instrument:

1. Visually confirm that the instrument is free and clear of any patient anatomy
2. If needed, request console surgeon to straighten instrument wrist and slightly open jaws
3. Use read-back technique to confirm which instrument to remove. Identify the instrument by both name and arm number.
4. Confirm that you are ready for the instrument to be removed
5. Squeeze release buttons on instrument housing and pull the instrument straight up and out without using excessive force to prevent instrument damage and patient harm

Warning: Be aware of the risk of damaging the drapes during instrument removal, to prevent a critical patient infection.

Warning: Insert and exchange instruments under vision to prevent tissue injury.

Note: During intraoperative use, do not clean tissue off of instruments with other instruments. Have the patient-side assistant remove the instrument and clean off excess debris with moist sterile gauze as necessary.

Guided Tool Change activity

Guided Tool Change is a feature that enables the system to remember the location of the last-installed instrument or endoscope on an arm.

1. Re-install an instrument onto the patient cart and notice the blinking green LEDs
2. The touchscreen will also show a message that states "Advance instrument to return to previous location" above the arm on which Guided Tool Change is active
3. Observing the monitor, and without pressing the instrument clutch button, push the instrument's housing to slowly guide the installed instrument through the cannula until you reach a hard stop, 3 mm short of the previous instrument tip location
   - Do not use excessive force during insertion to prevent patient tissue damage
   - Pressing any clutch button will clear memory and disable Guided Tool Change
   - In the absence of Guided Tool Change, always insert instruments under direct visualization of the endoscope
4. Practice using Guided Tool Change for endoscope-to-endoscope exchanges or endoscope removal and re-insertion to manage intra-operative fogging
5. Repeat steps 1-3 and demonstrate Guided Tool Change with a straight wrist and a bent wrist
6. If the system does not recognize a re-installed instrument, the arm is temporarily disabled and a message appears
   a. Remove the instrument, re-seat the sterile adapter and re-install the instrument
   b. If the issue persists, contact customer service

Port hopping activity

The endoscope or an instrument can be installed on any of the 4 arms as needed.

Note: Pay attention to arm associations and instrument control when the endoscope is port-hopped to prevent inadvertent instrument movement or removal, which could lead to tissue injury

1. Based on the activities completed in this section, install instruments in the configuration below, under direct visualization:
   a. Arm 1 – Bipolar instrument
   b. Arm 2 – Endoscope
   c. Arm 3 – Monopolar instrument
   d. Arm 4 – Grasper
2. Remove the bipolar instrument and grasper from arms 1 and 4 respectively
3. Use Guided Tool Change to install and insert the grasper in arm 1
4. Remove the monopolar from arm 3 and perform a Guided Tool Change to install it on arm 4
5. Remove the endoscope from arm 2 and install the endoscope on arm 3 (no Guided Tool Change as arm 3 previously had an instrument)
6. Under visualization, install and insert the bipolar on arm 2, guiding it towards the target anatomy (no Guided Tool Change as arm 2 previously had an endoscope)
7. Perform port hopping again, restoring instruments back to their original locations while verbalizing which ones will have Guided Tool Change and which will not

Instrument activation discussion

Observe hover and fire indications on the vision cart touchscreen associated with energy activation from the console:

- Hovering indications:
  - Instrument activation status area outline at the bottom of the screen will highlight in green
- Fire indications:
  - When pushing the primary blue pedal the entire box will highlight blue, the active function will be the only text displayed and an audible tone will be heard
  - When pushing the secondary yellow pedal the entire box will highlight yellow, the active function will be the only text displayed and an audible tone will be heard
- Non-energy instruments will have a black instrument activation status area

Learning check

1. What are the best practices for managing endoscope fogging?
2. What is Guided Tool Change?

10 Troubleshooting

Facilitator note To facilitate the below activity, position the patient cart arms close and facing one another, to trigger external and internal arm interferences.

Learning objectives

- Troubleshoot arm interferences and surgical workspace reach during setup and intraoperative use
- Describe emergency stop, manual instrument grip release and conversion processes, and the conditions under which these are performed
- Interpret vision cart touchscreen interface and patient cart LEDs to manage intraoperative system and generator faults

Arm and Instrument Management

Intraoperative arm management activity

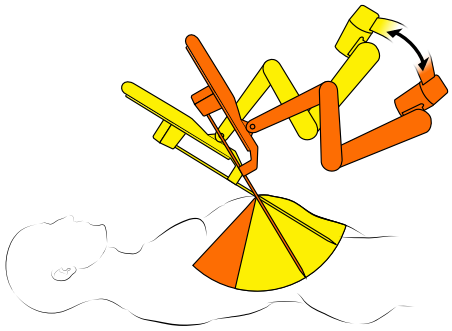
Resolve external interferences between the patient cart arms:

- Adjust flex joint to manage spacing between the front ends of the arms
  1. Remove the instruments from the arms that will be adjusted
  2. Stabilize the cannula while you reposition the arm using the port clutch button
  3. Reinsert the instrument manually under direct visualization
- Adjust patient clearance joint to manage spacing between the back ends of the arms
  1. Have the console surgeon move the tip of the instrument away from tissue
  2. Adjust the patient clearance joint up or down to provide additional spacing
  3. Note that instrument control from the surgeon console is paused when the patient clearance joint is being adjusted
- Create space between patient and patient cart arms by adjusting the patient clearance joint up

Increase instrument reach activity

Increase reach lateral to the surgical workspace (in line with ports)

1. Starting with the arm closest to the lateral workspace, flex all arms toward the direction where reach is desired, maintaining them parallel
2. When working away from this lateral region, consider flexing the arms to the center of the workspace for greater reach in the rest of the workspace



Increase reach beyond the level of ports

1. Have the console surgeon move the relevant instrument tip away from tissue
2. Adjust the patient clearance joint down to gain additional reach beyond the level of ports

The flex and patient clearance joints are adjusted to increase reach in the desired area during initial setup as well as during intraoperative use of the da Vinci Xi® system. Before adjusting flex joints, remember to remove instruments as this adjustment impacts the ports.

Fault Management

System faults discussion

When a fault occurs, the system determines whether the fault is recoverable or non-recoverable. The system behaves in the following manner when a fault occurs:

- Locks all the patient cart arms
  - In this state the patient cart clutch buttons function normally but the instrument is slightly harder to move and is not able to be left in clutch mode
- Sounds a series of error beeps
- Displays a text message on the monitors to describe the error
- If the fault is arm-specific, the system displays an error icon and the specific arm LED will be amber or red
- If the fault is not arm-specific, all arm LEDs will be amber or red

Recoverable fault activity

A recoverable fault is one that the system can recover from and does not require a system restart in order to resolve the fault. Amber LEDs on the arm/system are associated with a recoverable fault and the fault can be overridden by pressing the “Recover Fault” button. Follow the steps below to prevent patient harm.

1. Trigger a recoverable fault by pressing the Emergency Stop button on the patient cart helm or the surgeon console right side pod
2. Read messages on the vision cart touchscreen Fault tab and observe amber LEDs/audible feedback
3. Once the issue is resolved, tap the “Resume Use” button

Note If the fault is arm-specific, the system displays an error icon above that arm on the touchscreen and the specific arm LED will be red or amber depending on the type of fault. A “Disable Arm” option is displayed on the touchscreen and Surgeon Console touchpad. A disabled arm cannot be re-enabled until the system is restarted.

Non-recoverable fault discussion

A non-recoverable system fault is associated with red arm/system LEDs and requires a system restart.

Assuming non-recoverable fault, complete the below troubleshooting steps to prevent patient harm:

1. Read messages on surgeon console touchpad and vision cart touchscreen
2. Where possible, leave the instruments and endoscopes installed and inserted in the patient
3. Press the Power button on any system component to power off the system
4. Once powered off, wait for all system power buttons to be lit amber, indicating standby mode
5. Press the Power button on any system component to restart the system

Note During system restart, video is temporarily unavailable at the Surgeon Console viewer and touchscreen monitor.

Manual Instrument Grip Release

Instrument release kit activity

In the event of a system fault where normal instrument removal is not possible, or when surgeon console control of the instrument is not practical or available, use the instrument release kit (IRK).

- The instrument release kit can be individually sterile-wrapped, labeled, and placed in the vision cart drawer for easy access.

If the instrument tips are holding tissue, the instrument release wrench allows the patient-side operator to manually open the grips to release tissue. An instruction card is attached to wrench. Note that the instrument should not be re-used after the instrument release wrench has been applied.

1. To prevent patient injury, ensure that the system is in a fault state. If not, press the emergency STOP button to trigger a system recoverable fault.
2. Insert the long end of the wrench into the grip release socket on the instrument housing. Using the short end could lead to over-rotating which can cause unintended instrument motion or damage to the manual grip release mechanism, which could lead to tissue injury.
3. Stabilize the housing with one hand while pushing the wrench into the socket to ensure it engages to prevent patient injury.
4. Turn the wrench in the direction of the arrow on the housing approximately 1/4th turn to prevent unintended instrument motion or damage to the manual grip release mechanism, which could lead to tissue injury.
5. Under endoscopic visualization, verify the grips are not grasping tissue.
6. Once the tissue is released from the grips, remove the wrench from the instrument.
7. Squeeze the release buttons on the sides of the instrument housing and remove instrument.
8. Tap “Recover Fault” or “Resume Use” on the touchscreen or touchpad, restart the system as necessary.
9. Notify Technical Support of the issue. Provide event error code following these steps:
    a. On the vision cart touchscreen, touch Settings > Troubleshooting > Display Event Logs.
    b. The system log appears on the touchscreen. Use the navigation buttons to scroll through the log. Press “Close” to close the log.

Warning: Improper use of the Instrument Release Kit may cause injury to the patient or tissue.

Troubleshooting

Conversion and Power Loss

Conversion to open and lap/immediate patient access activity

If a situation arises where conversion to open surgery or immediate access to the patient is required:

1. Remove the instruments from the patient under endoscopic or direct visualization where possible
   a. If needed, use surgeon console control to release the instrument grips
   b. If robotic control cannot be used to release the instrument grips, manually release grips using the instrument release kit to prevent tissue damage when removing an instrument grasped on tissue
2. Remove the endoscope from the patient
3. Disconnect the cannulas from the arms
4. Move the arms away from the patient - If necessary, move patient cart away from the patient to prevent patient harm

Power loss situation discussion

- Battery level indicator at patient cart helm: number of green LEDs reduces as level reduces
- If the LED is red, it indicates that ≤ 10% battery is available and that the battery must be charged immediately
- In a power loss situation where the backup battery is depleted, you may need to manually raise the patient cart stabilization feet and place the patient cart in manual drive, in order to move it away from the patient

Power loss situation activity

1. Open the access panel in the front of the patient cart base.
2. Follow the instructions in the label inside the panel to place the patient cart into manual drive.
3. Make sure that the patient cart is not docked to the patient. If required during power loss, undock the arms and gently move the arms up along their vertical joints, overpowering the resistance to move the arms up and away from the patient.
4. Use two people to manually move the patient cart.
5. As soon as manual control is no longer needed, move the lever back to the powered position. This restores normal functionality of the system.



Learning check

1. Describe system behavior during a system fault
2. Describe the steps to use the instrument release kit

11 End of Procedure

Learning objectives

- Practice the processes for preparing the instruments, endoscopes and accessories for cleaning and sterilization
- Practice process of undraping the patient cart and preparing the system for shutdown
- Demonstrate shutdown of the system

Undocking the system discussion

Practice undocking the patient cart following the steps below to prevent patient injury.

1. Visually confirm that the instruments are free and clear of any patient anatomy
2. Remove the EndoWrist® instruments from the patient cart arms under direct visualization
3. Remove the endoscope from the patient cart arm
4. Detach the cannulas from the patient cart arms by pushing back the lever and detaching the cannula fin
5. Raise all the arms so they safely clear the patient and compact them as necessary to facilitate roll back
   a. If additional clearance is needed, use "extended port clutch" to raise the arm group
6. Check the room and around the patient for clearance
7. Carefully back up the patient cart away from the patient

Undraping the patient cart activity

1. For each arm, start at the front of the arm and release the instrument sterile adapter
2. Release the cannula mount sterile adapter
3. Release the arm clip on the gray handle near the instrument clutch button
4. Release the magnets at the back of the arm
5. From the back of each arm, roll the drape to the front, inverting it to trap debris internally
6. Dispose of the drape in a biohazard bin
7. With clean gloves (or assistance from a staff member with clean gloves), move the undraped arm away
8. Release the magnet at the top of the column drape
9. Remove the column drape
10. Dispose of the column drape a biohazard bin

End of procedure

Instrument point-of-use preparation discussion

1. Dispose of soiled drapes, cannula seals, expired, and single-use items in a biohazard waste bin.
2. Perform the following for instruments upon completion of the case:
    a. Remove disposable accessories, including disposable tips from the instruments (MCS tip cover).
    b. Keep the instrument tips moist intraoperatively and postoperatively before reprocessing.
    c. Check maximum use indicator – this is an indicator on the instrument housing that turns red when the instrument has reached its maximum number of uses.
    d. Wipe off excess soil.
3. Prime and wet:
    a. Prime: Insert the Luer tip fully into the primary flush port and twist to secure. Inject at least 15 mL of pH-neutral enzymatic cleaner or cold water into the port.
    b. Wet: Place the instruments in a bin with solution, or spray to wet all surfaces, and/or wrap a moistened cloth to keep the tip moist. Do not soak in saline.
4. Transport safely to decon – decon, or decontamination, is an area in hospitals and other healthcare facilities where medical devices are cleaned and disinfected. Decon is sometimes referred to as the “dirty side”. It is part of the sterile processing department (SPD) which can also be called the central sterile supply department (CSSD).
5. Reprocessing the instruments requires cleaning, disinfection, and sterilization. The "prime and wet" step for the instrument should begin immediately after the procedure to prevent drying. If prime and wet in the operating room is not possible within 60 minutes, start the prime and soak in decon within 60 minutes.

Endoscope point-of-use preparation discussion

Perform the following upon completion of the case:

1. Clean the cover (Proceed to step 2 if endoscope has white connector version VER -61 or higher)
    a. Clean the cover by wiping the end of the connector and inner surfaces of the connector cover with a disinfectant wipe or cloth sprayed with disinfectant. Follow disinfectant manufacturers’ instructions. Do not exceed a total contact time of 30 minutes.
    b. Allow the connector cover to dry fully.
    c. Place the connector cover on the connector and tighten the screws until secure. Do not over-tighten.
2. Wipe off excess soil.
3. Prime and wet the endoscope.
    a. Pull the release tab to open up the flush port cover and expose the 3 button flush ports.
    b. Find the button flush ports with buttons above.
    c. Prime and push buttons. Insert the Luer tip fully into each button flush port, and twist to secure. While pushing the corresponding button, inject at least 15 mL of pH-neutral cleaner or water into each of the 3 button ports.
    d. Place the endoscope in a bin with the solution, or spray to wet all surfaces, and/or wrap a moistened cloth to keep the tip moist. Do not soak in saline.

End of procedure

Endoscope point-of-use preparation discussion continued

4. Pack the endoscope (If connector cover is present, confirm it is secured to the connector).
   a. Place the endoscope in position in the appropriate tray.
      - Use validated sterilization trays
   b. Coil the cable.
   c. Place the connector in the correct position for the tray.
   d. Place the lid on the tray and ensure the cable is not pinched. Lock the lid latches.
5. Transport the endoscope to decon in sterilization tray that fully protects it, particularly the tip. Handle the endoscope with care at all times.
6. Reprocessing the endoscope requires cleaning, disinfection (optional), and sterilization. The "Prime and Wet" step for the endoscope should begin immediately after the procedure to prevent drying. If prime and wet in the operating room is not possible within 60 minutes, start the prime and soak in decon within 60 minutes.

Shutdown activity

1. Prepare patient cart for storage by stowing the system from the patient cart helm touchpad.
2. Check inventory from the touchscreen under the Settings tab.
3. Power off the ERBE VIO® dV generator.
4. Press the Power button on any system component to power the entire system off.
   a. The system will begin a 10 second shutdown process. Press the power button during this 10 second countdown to cancel the shutdown process.
   b. If the system remains shut down > 10 minutes, a restart is recorded as a new procedure. Instrument lives will be used accordingly.
5. It is now safe to disconnect all system cables.
   a. If possible, the system cables should not be disconnected. Keeping the cables connected will minimize exposure to contaminants.
   b. Store the system cables on each component's cable hook.
6. Remove non-system connections as necessary (auxiliary video and audio cables, ESUs, insufflators, etc.).

Learning check

1. What is the best practice of preparing the instruments for reprocessing?
2. What 3 features are necessary for procedure and endoscope trays?

12 Best Practices for da Vinci Surgical System Implementation

Leadership

- Define the leadership structure within the da Vinci® operating room to reduce variability between cases
- Identify core team participants (including patient-side assistant and anesthesiologist), emphasizing the importance
  of standardization
- Designate each role's territories of responsibility

  Example:

  Scrub: Sets up the back table and drapes the patient cart

  Circulator: Responsible for getting the patient

  Anesthesiologist: Induces and intubates the patient

Communication

- Emphasize the importance of two-way verbal communication between OR staff and surgeon
- Review "read-back" technique

  Example:

  Surgeon: "Please remove the monopolar curved scissors from arm 1 and replace with a large needle driver."

  Patient-side assistant: "Monopolar curved scissors coming out of arm 1. Large needle driver coming in."

  Surgeon: "Thank you."

Task assignment

- Each team member should be assigned a specific task to encourage task overlap
- Encourage team members to take initiative when they see a task that needs to be completed
- Emphasize that task overlap has the potential to maximize efficiency

Standardization

- Standardize the da Vinci system location, operating room layout, instrument trays, and core da Vinci team members to minimize variability between procedures
- Designate instrument and accessory selections for each procedure
- Confirm pick lists have been submitted to sterile processing

13 Da Vinci Case Walkthrough

OR staff only

OR Setup, Connections & Start Up	Draping	Port Placement	Docking	ERBE VIO dV Generator	Intraoperative Use	Trouble-shooting	End of Procedure
System connections	Drape column	Place ports	Deploy boom	Energy instrument setup	Instrument and endoscope insertion	Arm management	Undock
System startup	Drape arms	Position patient	Position patient cart		Instrument and endoscope removal	Fault management	Undrape
			Targeting		Instrument and endoscope exchange	Conversion	Point-of-use preparation
			Dock remaining arms and spacing		Intraoperative cleaning		Shutdown



Da Vinci technology training pathway

Feedback

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http://bit.ly/XiInService

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